

Task 1 — Project Report

Build & Evaluate a Linear Regression Model

Dataset	California Housing Dataset (scikit-learn)
Model	Linear Regression
Language	Python 3.14
Libraries	pandas, numpy, scikit-learn, matplotlib, seaborn
Date	June 2026

1. Objective

This project introduces the complete Machine Learning workflow by building and evaluating a Linear Regression model on the California Housing dataset. The goal is to predict median house values based on eight housing features, covering data loading, exploratory analysis, preprocessing, model training, evaluation, and result reporting.

2. Dataset Overview

The California Housing dataset is built into scikit-learn and contains data derived from the 1990 U.S. Census. It has **20,640 samples** with **8 features** and no missing values.

Feature	Description
MedInc	Median income in block group
HouseAge	Median house age in block group
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average number of household members
Latitude	Block group latitude
Longitude	Block group longitude
MedHouseVal (target)	Median house value in \$100,000s

3. Exploratory Data Analysis (EDA)

Key findings from the EDA:

- No missing values found across all 20,640 records — the dataset is clean and ready to use.
- MedInc (median income) has the strongest positive correlation with house price ($r = 0.69$), confirming that wealthier neighborhoods have higher home values.
- Latitude and Longitude have moderate negative correlations, showing that geography (coastal vs inland areas) significantly affects prices.
- The target variable MedHouseVal is right-skewed, with most values between \$1 and \$3 (in \$100,000s) and a spike at \$5 due to data capping.
- AveRooms and AveBedrms are highly correlated with each other ($r = 0.85$), indicating multicollinearity.
- Population and AveOccup have near-zero correlations with price, meaning crowd density alone is not a strong predictor.

Figure 1: Correlation Heatmap & House Price Distribution

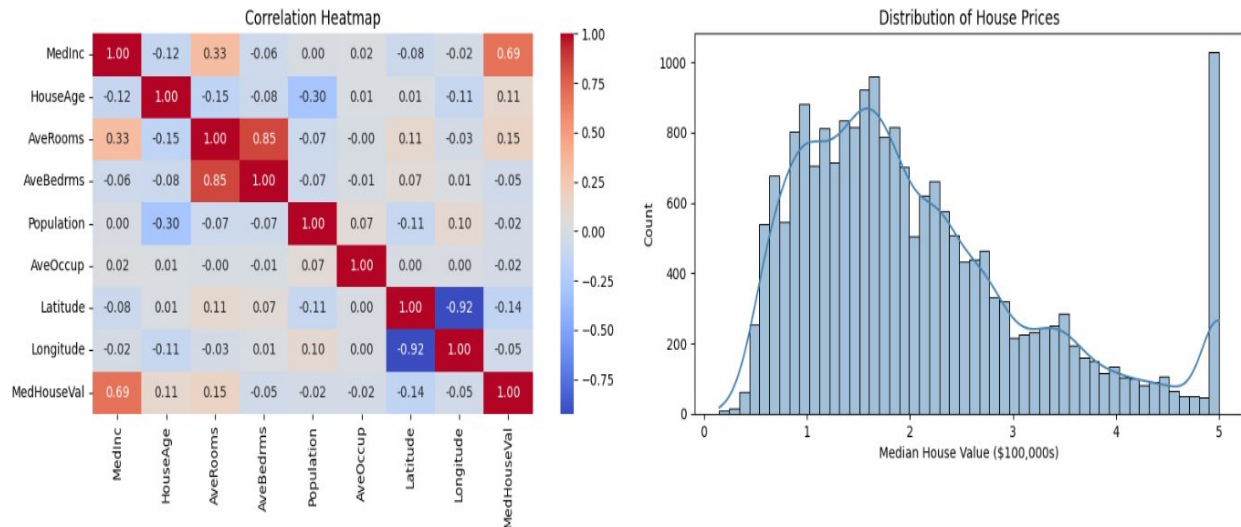


Figure 1: (Left) Correlation heatmap showing MedInc as the strongest predictor ($r=0.69$). (Right) Distribution of house prices — right-skewed with a cap at \$500,000.

4. Model — Linear Regression

Training Setup:

- All 8 features were used as input (X); MedHouseVal was the target (y).
- Data was split: 80% training (16,512 samples) and 20% testing (4,128 samples), using random_state=42 for reproducibility.
- Scikit-learn's LinearRegression() was trained on the training set and evaluated on the held-out test set.

Feature Coefficients (sorted):

Feature	Coefficient	Interpretation
AveBedrms	+0.7831	More bedrooms → higher price
MedInc	+0.4487	Higher income → higher price

HouseAge	+0.0097	Older homes slightly pricier
Population	~0.0000	Negligible effect
AveOccup	-0.0035	More occupants → slight decrease
AveRooms	-0.1233	Counterintuitive — likely due to multicollinearity
Latitude	-0.4198	Moving north → lower price
Longitude	-0.4337	Moving east → lower price

5. Model Evaluation & Results

Metric	Value	Meaning
MAE (Mean Absolute Error)	0.5332	Avg. prediction error of ~\$53,320
RMSE (Root Mean Squared Error)	0.7456	Penalizes larger errors more heavily
R ² (R-Squared Score)	0.5758	Model explains ~57.6% of variance

Interpretation: An R² of 0.5758 means the linear model captures about 58% of the variance in house prices. While acceptable as a baseline, the residual plot revealed non-random patterns, suggesting that the relationship between features and price is not purely linear.

Figure 2: Actual vs Predicted & Residual Plot

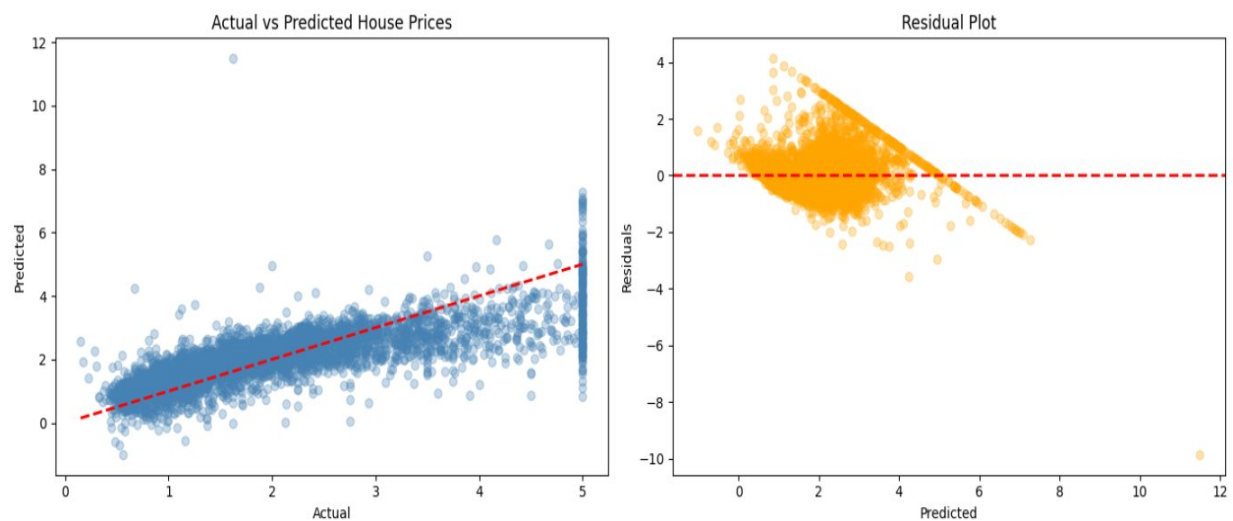


Figure 2: (Left) Actual vs Predicted scatter — points follow the diagonal trend line reasonably well. (Right) Residual plot — non-random pattern indicates heteroscedasticity, a limitation of linear regression.

6. Improvement Ideas

- **Feature Engineering:** Create new features like rooms-per-person ($\text{AveRooms} / \text{AveOccup}$) to capture richer relationships.

- **Log-transform target:** Apply $\log(\text{MedHouseVal})$ to reduce skewness and improve linearity.
- **Polynomial Features:** Use PolynomialFeatures to capture non-linear interactions between variables.
- **Regularization:** Try Ridge or Lasso regression to handle multicollinearity (AveRooms vs AveBedrms).
- **Advanced Models:** Use Random Forest or Gradient Boosting (XGBoost) which typically achieve $R^2 > 0.80$ on this dataset.
- **Feature Scaling:** Apply StandardScaler before training to normalize feature ranges.
- **Cross-Validation:** Use k-fold cross-validation instead of a single train/test split for more robust evaluation.

7. Conclusion

This project successfully implemented the complete ML pipeline — from data loading and EDA to model training, evaluation, and visualization. The Linear Regression baseline achieved an R^2 of 0.5758 and MAE of 0.5332, providing a solid starting point. The analysis clearly identifies where the model falls short (non-linearity, heteroscedasticity) and provides a clear roadmap for improvement. The trained model was saved as a pickle file for future use.